and a headset, sending data between a phone and a personal digital assistant, and printing from a laptop to a printer in the same room. For these scenarios, WiFi may be overkill in terms of power requirements and chip costs. Bluetooth is an unlicensed ad hoc 2.4 GHz standard for such applications. It has been slow to roll out, leading to speculation it would lose to WiFi. However, it now looks as though Bluetooth will find a niche, primarily replacing wires in short-range situations.

Unlicensed Metropolitan-Area Networking

At the other extreme from PANs are metropolitan-area networks (MANs), which cover entire neighborhoods or cities, though often designed primarily for business connections. The IEEE is developing standards, 802.16, for wireless MANs using the 10 GHz - 66 GHz spectrum. Though these are initially targeted at licensed spectrum, the same concepts could be applied in an unlicensed environment. Motorola offers a proprietary unlicensed system called Canopy that is designed for the metropolitan area. Canopy, or some variant of it, might become the basis for the 802.16 standard in unlicensed bands.

Lessons Of WiFi

The success of WiFi shows that spectrum sharing works in the real world. Without heavy-handed control by government or by service providers who have incentives to maximize only their own welfare, an entire industry has emerged. That industry has developed with no legal protections against competing uses. Despite repeated warnings of a "meltdown," only isolated anecdotal cases of congestion among WiFi users have been reported. Companies such as Intel and Microsoft are devoting substantial resources to these technologies, which they would be unlikely to do if they were seriously concerned about a tragedy of the commons.

Moreover, wireless LAN technology is evolving and diversifying rapidly. Vendors are beginning to deliver hybrid 802.11a/b chipsets, and devices that add software intelligence to WiFi are coming on the market. Innovation in the WiFi world follows the